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ENCLOSURES FOR PROVIDING A CONTROLLED ENVIRONMENT DURING THE PRODUCTION OF GLASS ARTICLES

Abstract

An enclosure for providing a controlled environment includes a central plane extending through a top end of the enclosure and a bottom end of the enclosure and bisecting the enclosure along a width of the enclosure, an inlet at the bottom end of the enclosure having an inlet width, W.sub.inlet, an enclosure wall extending from the inlet to the top end of the enclosure, an entry port at the top end of the enclosure, and an outlet between the entry port and the chamber region of the enclosure wall. The enclosure wall comprises a chamber region and a transition region between the inlet and the chamber region. The width of the chamber region, W.sub.chamber, is substantially constant through the chamber region. The width of the enclosure in the transition region decreases from W.sub.chamber to W.sub.inlet, and a ratio of W.sub.inlet to W.sub.chamber is from 1:2 to 1:5.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. patent application Ser. No. 17/410,131 filed on Aug. 24, 2021, which claims the benefit of priority under 35 U.S.C. § 119 of U.S. Provisional Application Ser. No. 63/071,570 filed on Aug. 28, 2020, the content of which are relied upon and incorporated herein by reference in their entirety.

TECHNICAL FIELD
[0002] Embodiments of the present disclosure generally relate to controlled environments, and, more specifically, to controlled environments for use in the production of glass articles.
BACKGROUND
[0003] Glass articles may be used in a variety of applications, including product packaging and specialty applications. Depending on the particular application, the glass article may include a coating on the exterior of the glass article to impart a particular characteristic, such as to reduce or prevent scratches, to reflect UV light, or to impart a color on the surface of the glass. Coatings may be applied during the glass manufacturing process using any one of a variety of known techniques. However, the performance of some techniques, such as spray coating techniques, often depends on the ambient environmental conditions in which the process occurs and in which the glass articles are located shortly after the process occurs. Moreover, vapors from solvents used in the coatings may evaporate from the surface of recently coated glass articles.
[0004] Accordingly, there is a need for enclosures to provide a controlled environment during the manufacture of glass articles with coatings.
SUMMARY
[0005] Various embodiments provide enclosures that surround the glass articles and provide a controlled environment during at least part of the glass manufacturing process. The enclosures described herein can reduce the amount of temperature-and-humidity-controlled, filtered air required for the environment, and efficiently capture solvent evaporating from the glass articles, thereby preventing the solvent vapors from contaminating the ambient environment. Additionally, the enclosures of various embodiments can reduce or prevent contamination of the glass article from particles present in the ambient environment that may render the glass article unsuitable for its intended end use application.
[0006] According to one or more embodiments, an enclosure for providing a controlled environment comprises a central plane extending through a top end of the enclosure and a bottom end of the enclosure and bisecting the enclosure along a width of the enclosure; an inlet at the bottom end of the enclosure having an inlet width, W.sub.inlet; an enclosure wall extending from the inlet to the top end of the enclosure; an entry port at the top end of the enclosure configured to receive a part carrier; and an outlet between the entry port and the chamber region of the enclosure wall. The enclosure wall comprises a chamber region and a transition region between the inlet and the chamber region. The width of the chamber region, W.sub.chamber, is substantially constant through the chamber region. The width of the enclosure in the transition region decreases from W.sub.chamber to W.sub.inlet, and a ratio of W.sub.inlet to W.sub.chamber is from 1:2 to 1:5. The central plane passes through the inlet and the entry port of the enclosure, and the outlet extends

along an outlet axis that is oriented at a non-zero angle with respect to the central plane.
[0007] According to one or more embodiments, a manufacturing line for producing glass articles comprises an enclosure, and a part carrier. The enclosure comprises a central plane extending through a top end of the enclosure and a bottom end of the enclosure and bisecting the enclosure along a width of the enclosure; an inlet at the bottom end of the enclosure having an inlet width, W.sub.inlet; an enclosure wall extending from the inlet to the top end of the enclosure; an entry port at the top end of the enclosure configured to receive a part carrier; and an outlet between the entry port and the chamber region of the enclosure wall. The enclosure wall comprises a chamber region and a transition region between the inlet and the chamber region. The width of the chamber region, W.sub.chamber, is substantially constant through the chamber region. The width of the enclosure in the transition region decreases from W.sub.chamber to W.sub.inlet, and a ratio of W.sub.inlet to W.sub.chamber is from 1:2 to 1:5. The central plane passes through the inlet and the entry port of the enclosure and the outlet extends along an outlet axis oriented at a non-zero angle with respect to the central plane. A gripping member of the part carrier is positioned through the entry port and the part carrier is configured to move a glass article through the chamber region of the enclosure.
[0008] According to one or more embodiments, a method for transporting a coated article is provided. The method comprises positioning the coated article within an enclosure; supplying a flow of fluid to the enclosure through the inlet; removing a flow of fluid from the enclosure through the outlet; and moving the coated article along a path through the enclosure, where the path is substantially parallel to the central plane. The enclosure comprises a central plane extending through a top end of the enclosure and a bottom end of the enclosure and bisecting the enclosure along a width of the enclosure; an inlet at the bottom end of the enclosure having an inlet width, W.sub.inlet; an enclosure wall extending from the inlet to the top end of the enclosure; an entry port at the top end of the enclosure configured to receive a part carrier; and an outlet between the entry port and the chamber region of the enclosure wall. The enclosure wall comprises a chamber region and a transition region between the inlet and the chamber region. A width of the chamber region, W chamber, is substantially constant through the chamber region. The width of the enclosure in the transition region decreases from W.sub.chamber to W.sub.inlet, and a ratio of W.sub.inlet to W.sub.chamber is from 1:2 to 1:5. The central plane passes through the inlet and the entry port of the enclosure and the outlet extends along an outlet axis oriented at a non-zero angle with respect to the central plane.
[0009] These embodiments are described in more detail in the following Detailed Description in conjunction with the appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0010] The following detailed description of specific embodiments of the present disclosure can be best understood when read in conjunction with the following drawings, where like structure is indicated with like reference numerals and in which:
[0011] FIG. **1** schematically depicts a cross section of an enclosure for providing a controlled environment, according to one or more embodiments disclosed herein;
[0012] FIG. **2** schematically depicts the flow of air through an enclosure for providing a controlled environment, according to one or more embodiments disclosed herein;
[0013] FIG. **3** schematically depicts a cross section of an enclosure with a glass article disposed therein, according to one or more embodiments disclosed herein;
[0014] FIG. **4** schematically depicts an alternate side view of an enclosure for producing glass articles, according to one or more embodiments disclosed herein;
[0015] FIG. **5** schematically depicts a side view of a manufacturing line for producing glass

articles, according to one or more embodiments disclosed herein;

[0016] FIG. **6** is a three-dimensional computational fluid dynamics (3D CFD) model of air flowing through an enclosure for providing a controlled environment, according to one or more embodiments disclosed herein;

[0017] FIG. **7** is a 3D CFD model of air flowing from a vial within an enclosure for providing a controlled environment, according to one or more embodiments disclosed herein.

[0018] FIG. **8** is a 3D CFD model of solvent evaporating from the surface of a vial and passing through an enclosure for providing a controlled environment according to one or more embodiments disclosed herein;

[0019] FIG. **9** is a 3D CFD model of air flowing through an enclosure for providing a controlled environment, according to one or more embodiments disclosed herein;

[0020] FIG. **10** is a 3D CFD model of particles flowing through an enclosure for providing a controlled environment, according to one or more embodiments disclosed herein; and

[0021] FIG. **11** is a 3D CFD model of particles flowing through an enclosure for providing a controlled environment, according to one or more embodiments described herein.

DETAILED DESCRIPTION

[0022] Reference will now be made in detail to embodiments of enclosures used in conjunction with the manufacture of coated glass articles, examples of which are illustrated in the accompanying drawings. Whenever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts. One embodiment of an enclosure is schematically depicted in FIG. **1**. The enclosure generally includes an enclosure wall, an inlet at the bottom end of the enclosure, a transition region, a chamber region, an entry port at the top of the enclosure and an outlet between the entry port and the chamber region. A central plane passes through the top end and the bottom end of the enclosure and bisects the enclosure along a width of the enclosure. Various embodiments of enclosures used for the manufacture of coated glass articles will be described herein with specific reference to the appended drawings.

[0023] It is noted that one or more of the claims presented herein utilize the term “wherein” as a transitional phrase. When used, this term is introduced in the claims as an open-ended transitional phrase that is used to introduce a recitation of a series of characteristics of the structure and should be interpreted in like manner as the more commonly used open-ended preamble term “comprising.”

[0024] Directional terms as used herein—for example up, down, right, left, front, back, top, bottom—are made only with reference to the figures as drawn and are not intended to imply absolute orientation.

[0025] Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

[0026] Unless otherwise expressly stated, it is in no way intended that any method set forth herein be construed as requiring that its steps be performed in a specific order, nor that with any apparatus specific orientations be required. Accordingly, where a method claim does not actually recite an order to be followed by its steps, or that any apparatus claim does not actually recite an order or orientation to individual components, or it is not otherwise specifically stated in the claims or description that the steps are to be limited to a specific order, or that a specific order or orientation to components of an apparatus is not recited, it is in no way intended that an order or orientation be inferred, in any respect. This holds for any possible non-express basis for interpretation, including: matters of logic with respect to arrangement of steps, operational flow, order of components, or orientation of components; plain meaning derived from grammatical organization or punctuation, and; the number or type of embodiments described in the specification.

[0027] As used herein, the singular forms "a," "an" and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to "a" component includes aspects having two or more such components, unless the context clearly indicates otherwise.
[0028] As used herein, a "controlled environment" refers to an enclosed or partially enclosed volume in which certain atmospheric conditions are maintained within set boundaries. For example, temperature, pressure, and humidity may be kept within specified ranges within a controlled environment. Additionally, the entrance of particles into the controlled environment may be regulated, for example, by filtration of incoming air.
[0029] As used herein, "laminar flow" refers to the flow of a fluid including, without limitation, air that is free from cross-currents, eddies, swirls, and lateral mixing.
[0030] As used herein, an "inflection point" refers to a point along a curve where the curve transitions from concave to convex or from convex to concave.
[0031] As used herein, "reflection symmetry" refers to symmetry of an object with respect to a plane wherein the reflection of the object along that plane is indistinguishable from the object itself.
[0032] FIG. **1** depicts an embodiment of an enclosure **100** for providing a controlled environment along at least a portion of, for example, a glass manufacturing line for producing coated glass articles. In particular, FIG. **1** schematically depicts a cross-section of the enclosure **100** along an XY plane as defined by the coordinate axes included in FIG. **1**. A central plane **101** provides a reference for several aspects of the enclosure **100**. Specifically, the central plane **101** provides a reference for defining the symmetry of the enclosure **100**, a position of an inlet **104** and an entry port **108** of the enclosure **100**, and an orientation of an outlet **109** of the enclosure **100**. In embodiments, the central plane **101** extends through a top end **102** and a bottom end **103** of the enclosure **100** in the +/−X direction. Additionally, the central plane **101** extends in the Z direction through the length of the enclosure. In embodiments, the central plane **101** bisects the enclosure **100** along a width of the enclosure, for example, along a width of the inlet (W.sub.inlet **110**), a width of a chamber region (W.sub.chamber **111**), or a width of an entry (W.sub.entry **113**).
[0033] The enclosure **100** includes at least one enclosure wall **105**, which extends from the bottom end **103** of the enclosure **100** to the top end **102** of the enclosure **100** and defines various regions of the enclosure **100**. For example, the enclosure wall **105** defines a chamber region **106** and a transition region **107** of the enclosure **100**. In some embodiments, the enclosure **100** includes two enclosure walls that oppose one another. For example, the enclosure **100** may include enclosure walls on opposite sides of the central plane **101**. In embodiments, the enclosure wall **105** includes one or more apertures through the enclosure wall **105**. For example, apertures through the enclosure wall **105** can include the inlet **104** and the outlet **109** of the enclosure **100**. In embodiments, the enclosure wall **105** may comprise any material that is sufficiently smooth to facilitate laminar flow of air through the enclosure. For example, the enclosure wall **105** may be formed from an opaque material, such as sheet metal or other, similar materials. In some embodiments, a transparent material, such as a polymer or plastic resin (e.g., PLEXIGLAS™ available from Arkema), may form the enclosure wall **105** to allow visual inspection of the various regions of the enclosure **100**, including the chamber region **106** and the transition region **107**. In further embodiments, the enclosure wall **105** may be formed from a combination of opaque and transparent materials, such that at least a portion of the enclosure's interior may be visible from outside the enclosure **100**. For example, in such embodiments, at least a portion of the chamber region **106** or the transition region **107**, or both may be visible from the exterior of the enclosure **100**.
[0034] The inlet **104** is located at the bottom end **103** of the enclosure **100**. In embodiments, the inlet **104** is an opening in the enclosure wall **105** that allows air to enter the enclosure **100**. Alternatively, the inlet **104** may be formed between two opposing enclosure walls instead of being formed through a single enclosure wall. The inlet **104** has a width, W.sub.inlet **110**. In FIG. **1**, the width is measured in the Y direction from the inner surface **155** of the enclosure wall **105** to an

opposing inner enclosure wall surface (e.g., the width is an interior width of the inlet **104**). In embodiments, W.sub.inlet **110** is greater than or equal to 4 mm and less than or equal to 45 mm. For example, W.sub.inlet **110** may be from 4 mm to 45 mm, from 4 mm to 40 mm, from 4 mm to 35 mm, from 4 mm to 30 mm, from 4 mm to 25 mm, from 4 mm to 20 mm, from 4 mm to 15 mm, from 4 mm to 10 mm, from 10 mm to 45 mm, from 15 mm to 45 mm, from 20 mm to 45 mm, from 25 mm to 45 mm, from 30 mm to 45 mm, from 35 mm to 45 mm, or even from 40 to 45 mm. In embodiments, the central plane **101** passes through the inlet **104** and bisects W.sub.inlet. In such embodiments, W.sub.inlet **110** is measured normal to the central plane **101**. Although depicted in FIG. **1** as having a constant width over a height (+/−X direction) of the inlet **104**, it is contemplated that in embodiments, the width of the inlet **104** may vary. In such embodiments, W.sub.inlet corresponds to the minimum interior width of the inlet **104**.

[0035] As shown in FIG. **1**, the enclosure wall **105** of the enclosure **100** defines a chamber region **106**. In embodiments, the enclosure wall **105** extends in the X and Z directions through the chamber region **106** and is substantially parallel to the central plane **101** through the chamber region **106**.

[0036] The chamber region **106** has a width, W.sub.chamber **111**. In FIG. **1**, the width is measured in the Y direction from the inner surface **155** of the enclosure wall **105** to an opposing inner surface **155** of the enclosure wall **105** (e.g., the width is an interior width of the chamber region **106**). In embodiments, W.sub.chamber **111** is substantially constant throughout the chamber region **106**. In embodiments, W.sub.chamber **111** is greater than or equal to 20 mm and less than or equal to 90 mm. For example, W.sub.chamber **111** may be from 20 mm to 90 mm, from 30 mm to 90 mm, from 40 mm to 90 mm, from 50 mm to 90 mm, from 60 mm to 90 mm, from 70 mm to 90 mm, or even from 80 mm to 90 mm. In further examples, W.sub.chamber **111** may be from 20 mm to 80 mm, from 20 mm to 70 mm, from 20 mm to 60 mm, from 20 mm to 50 mm, from 20 mm to 40 mm, or even from 20 mm to 30 mm. Other values for the width **111** of the chamber region **106** are possible, provided that the width **111** is large enough to allow for adequate airflow around the maximum diameter of the part being transported through the enclosure **100**. In embodiments, W.sub.chamber **111** is from 2 to 3 times the maximum diameter of the part. Additionally, in embodiments, the central plane **101** passes through the chamber region **106** of the enclosure **100** and bisects W.sub.chamber **111**. In such embodiments, W.sub.chamber **111** is measured normal to the central plane **101**.

[0037] The enclosure **100** further comprises a transition region **107** between the inlet **104** and the chamber region **106**. In embodiments, the transition region **107** is enclosed by a section of the enclosure wall **105** that is not parallel to the central plane **101**. In the transition region **107**, the width (e.g., the interior width) of the enclosure **100** decreases from W.sub.chamber **111** to W.sub.inlet **104**. In embodiments, the decrease in width of the enclosure **100** takes place over a distance **112** (e.g., measured in the +/−X direction in FIG. **1**) parallel to the central plane **101**. In embodiments, the distance **112** is greater than or equal to 200 mm and less than or equal to 900 mm. For example, the distance **112** may be from 200 mm to 900 mm, from 300 mm to 900 mm, from 400 mm to 900 mm, from 500 mm to 900 mm, from 600 mm to 900 mm, from 700 mm to 900 mm, or even from 800 mm to 900 mm. In further examples, the distance **112** may be from 200 mm to 800 mm, from 200 mm to 700 mm, from 200 mm to 600 mm, from 200 mm to 500 mm, form 200 mm to 400 mm, or even from 200 mm to 300 mm. In further embodiments, the distance **112** may be about ten times larger than W.sub.chamber **111**. Other distances are possible and contemplated, provided that one-dimensional fluid flow is maintained at both the chamber-end and the inlet end of the transition region **107**. In embodiments, the ratio of W.sub.inlet **110** to W.sub.chamber **111** is from 1:2 to 1:5 or from 1:3 to 1:4. Without being bound by theory, it is believed that a transition region **107** dimensioned as described above (i.e., where the ratio of W.sub.inlet **110** to W.sub.chamber **111** is from 1:2 to 1:5) may facilitate laminar air flow through the enclosure **100** by ensuring that the change in width of the enclosure **100** is not so sudden that eddies or other turbulent flow patterns are present in the air flowing from the inlet **104** into the chamber region **106**. However, it is contemplated that in embodiments, the flow through the transition region **107** may include transitional or turbulent flow, although the flow of fluid at both the chamber-end and the inlet end of the transition region **107** is one-dimensional.

[0038] In embodiments, the enclosure wall **105** has an S-shaped curve within the transition region **107** of the enclosure **100**, as illustrated in FIG. **1**. In embodiments, the S-shaped curve comprises an inflection point **157** at which a change in a direction of the curvature occurs. Without being bound by theory, it is believed that the presence of an S-shaped curve in the transition region **107** may help facilitate laminar air flow through the enclosure **100** by providing a smooth transition from the inlet **104** to the chamber region **106** of the enclosure **100**. Although an S-shaped curve is specifically shown and described herein, it should be understood that other smooth curves and transitional shapes are possible and contemplated, provided they do not cause air flowing through the enclosure **100** to recirculate or become turbulent upon entry into the chamber region **106**. For example, in embodiments, the transition region **107** can be linear given a large enough distance **112** to enable one-dimensional fluid flow to be preserved without backflow.

[0039] The enclosure **100** also comprises an entry port **108** at the top of the enclosure **100** that is configured to receive a part carrier (not shown in FIG. **1**). The entry port **108** can be, for example an opening or slot through which the part is conveyed. The entry port **108** has a width, W.sub.entry **113**, measured in the Y direction from an inner surface **158** of the entry port **108** to an opposing inner surface **158** of the entry port **108**. In embodiments, W.sub.entry **113** is contingent on the dimensions of the part carrier. In embodiments, W.sub.entry **113** is greater than or equal to 1.0 cm and less than or equal to 5.0 cm. For example, W.sub.entry **113** may be from 1.0 cm to 5.0 cm, from 1.5 cm to 5.0 cm, from 2.0 cm to 5.0 cm, from 2.5 cm to 5.0 cm, from 3.0 cm to 5.0 cm, from 3.5 cm to 5.0 cm, from 4.0 cm to 5.0 cm, or even from 4.5 cm to 5.0 cm. In further examples, W.sub.entry **113** may be from 1.0 cm to 4.5 cm, from 1.0 cm to 4.0 cm, from 1.0 cm to 3.5 cm, from 1.0 cm to 3.0 cm, from 1.0 cm to 2.5 cm, from 1.0 cm to 2.0 cm, or even from 1.0 cm to 1.5 cm. However, it should be understood that other dimensions for W.sub.entry **113** are contemplated and possible, provided that the part carrier can freely move through the entry port **108** (e.g., along the +/−Z direction in FIG. **1**). In embodiments, W.sub.entry **113** is less than W.sub.chamber **111**. As shown in FIG. **1**, the central plane **101** passes through the entry port **108** and, in embodiments, bisects the entry port **108** along W.sub.entry **113**. In such embodiments, W.sub.entry **113** is measured normal to the central plane.

[0040] In embodiments, the enclosure **100** further comprises at least one outlet **109** positioned between the entry port **108** and the chamber region **106** of the enclosure **100**. In embodiments, the width of the outlet **109** is from 0.5 cm to 3.0 cm. For example, the width of outlet **109** may be from 0.5 cm to 3.0 cm, from 1.0 cm to 3.0 cm, from 1.5 cm to 3.0 cm, from 2.0 cm to 3.0 cm, or even from 2.5 cm to 3.0 cm. In further examples, the width of outlet **109** may be from 0.5 cm to 2.5 cm, from 0.5 cm to 2.0 cm, from 0.5 cm to 1.5 cm, or even from 0.5 cm to 1.0 cm. The outlet **109** extends along an outlet axis **112**. In embodiments, the outlet axis **112** is oriented at a non-zero angle with respect to the central plane **101**. For example, the outlet axis **112** may be normal with respect to the central plane **101**. In the embodiment shown in FIG. **1**, the outlet axis **112** extends from the central plane **101** in the +/−Y direction and lies within a YZ plane. In some embodiments, the enclosure **100** may comprise two outlets **109** where the outlets **109** are on opposite sides of the central plane **101** and the outlets **109** extend away from the central plane **101** along the outlet axis **112**. For example, in embodiments in which the left wall and right wall of the enclosure **100** are different walls separated from one another at the ends of the enclosure **100**, each wall can include an outlet **109**, with the outlets being symmetrically situated. As another example, in embodiments in which the "left wall" and the "right wall" of the enclosure are different regions along a length of material (e.g., where a single wall has a circular or oval-shape), the outlet **109** can be a single channel running along the length of the wall.

[0041] In various embodiments, the enclosure **100** comprises reflection symmetry with respect to the central plane **101**. Without wishing to be bound by theory, it is believed that the symmetry of the enclosure **100** may facilitate a symmetric airflow pattern through the enclosure **100**, which, in turn, may facilitate smooth, or laminar, airflow through the enclosure **100**.

[0042] Turning now to FIG. **2**, fluid flow through the enclosure **100** is shown. The fluid flow is generally represented by flow lines **201-204**. It should be understood that the flow lines **201-204** depict exemplary paths that fluid, such as air, may take through the enclosure **100** and that the flow of fluid is not limited specifically to the flow lines **201-204** in embodiments described herein.

[0043] Fluid enters the enclosure **100** through the inlet **104** as indicated by flow lines **201**. The fluid is supplied to the inlet **104** at a predetermined temperature and humidity from a fluid source, such as fluid source **503** (FIG. **5**). In embodiments, the fluid has a temperature of greater than or equal to 20° C. and less than or equal to 25° C. and a relative humidity (RH) of less than 60%. For example, the fluid may be supplied at a temperature from 20° C. to 25° C., from 21° C. to 25° C., from 22° C. to 25° C., from 23° C. to 25° C., or even from 24° C. to 25° C. In embodiments, the temperature is measured by any suitable means, for example, by thermometer, thermography, or the like. In further examples, the fluid may be supplied at a RH of less than 60%, less than 55%, less than 50%, less than 45%, or even less than 40%. In embodiments, RH is measured by any suitable means, for example, by hygrometer. Without being bound by theory, it is believed that having the glass articles in an environment at temperatures between 20° C. and 25° C. and at a RH below 60% shortly after coating may facilitate the evaporation of solvent used in the coating process and provide a uniform, high-quality coating on the glass articles.

[0044] In embodiments, the fluid supplied to the enclosure **100** is filtered before entering the enclosure **100**. This can be achieved by passing the fluid through a high efficiency particulate air (“HEPA”) filter before the fluid is provided to the inlet **104**. Any suitable HEPA filtration system known in the art may be used to filter the fluid entering the enclosure **100**. Moreover, it is contemplated that filters other than HEPA filters can be used, depending on the particular embodiment. Without being bound by theory, it is believed that passing the fluid through a HEPA filter may remove particles from the fluid which could interfere with the coating on glass articles within the enclosure. Thus, filtering the fluid before it enters the enclosure may lead to an increased coating quality on the glass articles. Additionally, passing the fluid entering the enclosure through a HEPA filter may remove particles that would contaminate the glass articles from entering the enclosure. Thus, filtering the fluid before it enters the enclosure may ensure that the glass articles are free from contaminants and suitable for use in, for example, pharmaceutical applications.

[0045] Referring again to FIG. **2**, the fluid supplied to the enclosure **100** through the inlet **104** travels along flow lines **201** through the transition region **107** of the enclosure **100**. In embodiments, fluid flow through the transition region **107** is substantially laminar (i.e., without eddies or other turbulent flow patterns). In such embodiments, the fluid passing from the inlet **104** through the transition region **107** moves toward the top end **102** of the enclosure **100** (e.g., in the +X direction), regardless of its motion in the Y and Z directions. In other words, the fluid flow through the transition region **107** is free from eddies or currents in which the air flows toward the bottom end **103** of the enclosure **100** (e.g., in the −X direction). Without wishing to be bound by theory, it is believed that the flow of fluid through the transitions region remains substantially laminar due to the smooth contours in the transition region as well as the ratio of W.sub.inlet **110** to W.sub.chamber **111**.

[0046] The flow lines **202** depicted in FIG. **2** depict the flow of fluid through the chamber region **106** of the enclosure **100**. As in the transition region **107**, the flow of fluid through the chamber region **106** is substantially laminar. As depicted by flow lines **202**, the fluid continues to move toward the top end **102** of the enclosure **100** (e.g., in the +X direction) until it reaches the outlet **109**. Once the fluid reaches the outlet **109**, the fluid flows through the outlet **109** and is removed from the enclosure **100**. To assist in maintaining the flow of fluid in the +X direction from the inlet

104 to the outlet **109**, in embodiments, a vacuum is applied to the outlet **109** to establish a pressure differential through the enclosure **100**, as will be discussed in greater detail herein.

[0047] In various embodiments, the fluid flow through the enclosure **100** is symmetric about the central plane **101**. As depicted in FIG. **3**, when a glass article **302**, such as a glass vial, is positioned in the enclosure **100**, the symmetric fluid flow uniformly engulfs the surfaces of the glass article **302** with conditioned fluid that enters the enclosure through the inlet **104**, which may ensure that the glass article **302** is protected from ambient air that could contain particles that could disrupt the coating or contaminate the glass article **302**. That is, the introduction of fluid into and through the enclosure **100** both flushes the volume of the enclosure of ambient air as well as supplies the interior volume of the enclosure with conditioned fluid (i.e., fluid having a desired temperature and/or relative humidity) which assists in uniformly processing a coating applied to the glass article. In embodiments, the glass article **302** may be a pharmaceutical container, such as a vial or syringe; however, the glass article **302** is not limited to such containers.

[0048] Returning to FIG. **2**, it is also contemplated that some amount of ambient air may enter the enclosure **100** through the entry port **108** along flow lines **203**. In contrast to the fluid supplied through the inlet **104**, the ambient air entering the enclosure **100** through the entry port **108** is not temperature- or humidity-controlled or filtered to remove particulates. Accordingly, in embodiments, the position of the outlet **109** near the top end **102** of the enclosure **100** mitigates the flow of ambient air into the chamber region **106** through the entry port **108**. As depicted by the flow lines **204**, the ambient air exits the enclosure **100** through the outlet **109** before it is able to proceed into the chamber region **106**.

[0049] As set forth above, in embodiments, the outlet **109** is fluidly coupled to a vacuum source (e.g., vacuum source **504** in FIG. **5**), which assists in maintaining the flow of fluid through the enclosure **100** as described above. In particular, the vacuum can help to ensure that the ambient air exits the enclosure **100** before entering the chamber region **106** and that the fluid supplied through the inlet **104** constantly moves toward the top end **102** of the enclosure **100**. Additionally, the vacuum may induce pressures within the enclosure **100** below ambient pressure, which, in embodiments, ensures that any solvent evaporating from the coatings on the glass articles within the enclosure **100** exits through the outlet **109**, as further discussed herein. Pressure within the enclosure **100** can be maintained at a pressure below ambient air pressure by controlling the vacuum fluidly coupled to the outlet **109** as well as the flow rate of fluid supplied to the enclosure **100** through the inlet **104**, so the flow rate of fluid supplied to the enclosure **100** through the inlet **104** is less than or equal to the flow rate of fluid exiting the enclosure **100** through the outlet **109**. Accordingly, an aggregate flow of fluid through the outlet **109** is greater than the flow of fluid entering the enclosure through the inlets to maintain a negative pressure within the chamber region **106**. Any suitable method for controlling the flow rate of fluid known and used in the art can be used to ensure that the pressure within the enclosure **100** is maintained below ambient air pressure, such as a vacuum pump.

[0050] Referring now to FIGS. **3** and **4**, the glass article **302** is coupled to a part carrier **300** through a gripping member **301** configured to move the glass article **302** along a manufacturing line through the enclosure **100**. In particular, the gripping member **301** of the part carrier **300** extends through the entry port **108** of the enclosure **100** and holds the glass article **302** within the chamber region **106** of the enclosure **100**. Suitable gripping members and part carriers are further described in U.S. Pat. No. 10,576,494, the entirety of which is incorporated by reference herein. The part carrier **300** moves the glass article **302** through the chamber region **106** of the enclosure **100** along a manufacturing line path (e.g., in the +Z direction), as depicted in FIGS. **3** and **4**. In embodiments, the part carrier **300** moves the glass article **302** through the chamber region **106** along the central plane **101**. In embodiments, the glass article **302** may be situated within the enclosure **100** such that the central plane **101** generally bisects the glass article.

[0051] In embodiments, the part carrier **300** comprises a plate **303** positioned between the entry port **108** and the chamber region **106** of the enclosure **100**. The gripping member **301** is attached to the plate **303** so the plate **303** moves with the gripping member **301** through the enclosure **100**. In embodiments, the plate **303** may be oriented such that a surface of the plate adjacent the entry port **108** lies in a plane normal to the central plane **101**.

[0052] The plate **303** has a width, W.sub.plate **304**. As used herein, W.sub.plate **304** refers to a maximum distance between two points on the edge of the plate **303** when measured across the plate **303** (as opposed to around the perimeter of the plate **303**). In embodiments, W.sub.plate **304** is greater than or equal to W.sub.chamber **111**. When W.sub.plate **304** is greater than W.sub.chamber **111**, the plate **303** extend into the outlet **109**. In embodiments, W.sub.plate **304** is greater than or equal to W.sub.entry **113**. In such embodiments, the plate **303** ensures that there is no direct, linear path between the entry port **108** and the glass article **302**. Accordingly, the plate **303** can deflect ambient air entering from the entry port **108** toward the outlet **109**. Additionally, any particles that enter the enclosure **100** through the entry port **108** can be intercepted by the plate **303** and prevented from contacting the glass articles **302**. In embodiments, W.sub.plate **304** may be greater than W.sub.entry **113** and less than W.sub.chamber **111**.

[0053] In embodiments, the plate **303** may be in the form of a circular disk, although other shapes are contemplated. When the plate **303** has a circular shape, W.sub.plate corresponds to a diameter of the plate **303**. In embodiments, a circular disk can advantageously uniformly direct the path of ambient air from the entry port **108** away from the glass article **302** being conveyed through the enclosure **100**. Additionally, in embodiments in which the gripping member **301** rotates within the enclosure **100** (e.g., about an axis extending in the +/−X direction in the FIGS), the use of a circular disk maintains a constant relationship between W.sub.plate and W.sub.chamber or W.sub.entry at any point during the rotation of the gripping member **301**.

[0054] FIG. **4** schematically depicts a side view of the enclosure **100** depicted in FIG. **3** to better illustrate the central plane **101** and the part carrier **300**. As can be seen in FIG. **4**, the part carrier **300** forms part of a manufacturing line for producing glass articles **302**. The central plane **101** is an XZ plane and extends through the inlet **104** at the bottom end **103** of the enclosure **100** and through the entry port **108** at the top end **102** of the enclosure **100**. Additionally, the central plane **101** extends along a length of the enclosure **100**, where the length of the enclosure is measured in the +/−Z direction.

[0055] As shown in FIG. **4**, the part carrier **300** includes a plurality of gripping members **301** that may be conveyed through the enclosure **100**. Each of the gripping members **301** is coupled to and moves a corresponding glass article **302** through the chamber region **106** of the enclosure **100**. The gripping members **301** move along a length of the enclosure **100** in series along the manufacturing line path. In one or more embodiments, the gripping members **301** move the glass articles **302** through the chamber region **106** along the central plane **101**.

[0056] Referring now to FIG. **5**, the part carrier **300** is positioned to facilitate movement of the glass articles **302** from a coating apparatus **501**, where a coating is applied to the glass articles **302**, through the enclosure **100**, and to a curing apparatus **502**, where the coating is cured on the surface of the glass articles **302**. Other locations of the part carrier **300** (and the enclosure **100**) within a glass article manufacturing line are contemplated and possible, depending on the particular embodiment. Moreover, it is contemplated that in embodiments, the gripping member **301** of the part carrier **300** can be replaced with a different interface for engaging the glass article **302**, such as a platform on which the glass article **302** is positioned, a suction apparatus, or the like.

[0057] With reference to FIG. **5**, in use, the gripping member **301** of the part carrier **300** engages a glass article **302**, such as by a vacuum chuck or closing robotic fingers around a neck region of the glass article **302**. The glass article **302** can be engaged, for example, in a coating apparatus **501** or upstream of the coating apparatus **501** (e.g., prior to the glass article **302** entering the coating apparatus **501**) as the glass article **302** moves along a glass article manufacturing line. The gripping member **301** moves along the manufacturing line path and enters the enclosure **100**, thereby traversing a glass article **302** into and through the enclosure **100**.

[0058] In embodiments, the enclosure **100** can include an open end (not shown) that enables the glass article **302** to be passed between enclosure walls **105** of the enclosure **100** such that the glass article **302** is positioned within the chamber region **106** and the gripping member **301** extends through the entry port **108**. In such embodiments, a fluid knife, such as an air knife or the like, may be positioned along the open end to prevent entry of particles and ambient air into the enclosure **100**. In embodiments in which the enclosure **100** includes a single enclosure wall **105** that curves at one or both ends of the enclosure **100** such that the enclosure wall **105** includes a first portion of the inner surface **155** of the enclosure wall **105** that is parallel to and faces a second portion of the inner surface **155** of the enclosure wall **105**, the entry port **108** can include a region near the end of the enclosure **100** that has a width that is wider than the glass article **302** and the part carrier **300** (including the gripping member **301** and the plate **303**). In such embodiments, the part carrier **300** can vertically lower the glass article **302** (e.g., move the glass article **302** in the +/−X direction) into the chamber region **106** of the enclosure **100** through the entry port **108** as it moves the glass article **302** along the manufacturing line path (e.g., along the +/−Z direction).

[0059] As shown in FIG. **5**, conditioned fluid enters the enclosure **100** through the inlet **104** from a fluid source **503**, which, in embodiments, is coupled to the inlet **104** by a manifold **510**. In embodiments, the manifold **510** may include baffles or perforated plates to facilitate uniform fluid flow through the inlet **104**. The fluid is conditioned to have a predetermined temperature, humidity, and particulate level, which can be achieved through various heating, humidity, and filtration systems. In embodiments, the conditioned fluid is provided to the inlet **104** at a predetermined flow rate and pressure, and flows through the enclosure **100** to the outlet **109**, as was previously described with reference to FIG. **2**.

[0060] As described above, in embodiments, a vacuum source **504** is additionally applied to the outlet **109** of the enclosure **100**. Accordingly, a vacuum pump or other vacuum source **504** can be fluidly coupled to the outlet **109** by a manifold **511** to pull fluid from the enclosure **100** through the outlet **109**. The vacuum source **504**, in embodiments, further establishes a negative pressure within the enclosure **100**, as described above, to facilitate flow of the conditioned fluid from the bottom end **103** of the enclosure **100** toward the top end **102** of the enclosure **100** and through the outlet **109**.

[0061] The part carrier **300** moves the glass articles **302** through the enclosure **100** along the manufacturing line path and, in embodiments, may further rotate the glass articles **302** around a rotation axis extending in the +/−X direction through the center of the glass article **302** and lying in the central plane **101**. In embodiments, the part carrier **300**, and more particularly, the gripping member **301**, rotate the glass articles **302** at a rate of from 1000 to 3000 rotations per minute (RPM).

[0062] In embodiments, when the glass article **302** enters the enclosure **100**, the glass article **302** has a coating thereon that includes one or more solvents. This solvent may evaporate from the surface of the glass article **302** while the coated glass article **302** is moving through the enclosure **100**, thereby forming solvent vapors within the enclosure **100**. For example, the solvent vapors may be released from the surface of the glass article **302** during a partial curing of the coating as a result of the temperature and/or humidity of the environment within the enclosure **100**. In embodiments, the solvent vapors are flushed from the enclosure **100** by the fluid and exit the enclosure **100** through the outlet **109**. Without being bound by theory, it is believed that maintaining a pressure below ambient pressure within the enclosure **100**, as discussed previously, may prevent the escape of solvent from the enclosure **100** into the atmosphere. Negative pressure within the enclosure **100** ensures that nearly all the fluid within the enclosure **100** exits the enclosure **100** through the outlet **109**, allowing the solvent to be removed from the fluid before the fluid is released into the environment.

[0063] In embodiments, fluid including the solvent vapors are directed from the outlet **109**, through the manifold **511** and the vacuum source **504**, and to a solvent recovery system **505** or an air remediation system that filters, adsorbs, or otherwise separates the solvent vapors from the fluid flowing through the outlet **109** before the fluid is released into the ambient environment or recycled. Such solvent capture may reduce the presence of solvent in the ambient environment and also facilitate reclamation and recycling of the fluid and solvent.

[0064] In embodiments, the enclosure **100** may be temperature controlled such that the enclosure **100** may act as a curing chamber. In such embodiments, the temperature within the enclosure is kept above 300° C. or higher, depending on the curing temperature of the coating. Accordingly, in embodiments, the air entering the enclosure **100** is heated. For example, heaters on the outside of metal pathways supplying air to the enclosure **100** may bring the temperature of the air entering the enclosure above 300° C. Alternatively, the air may be passed through a heating unit to bring the air to the desired temperature. Using the enclosure **100** as a curing chamber may prevent particles from sticking to a coating on the glass article before the coating is cured. Additionally or alternatively, in embodiments, the enclosure **100** is temperature controlled to control solvent flash-off. In such embodiments, the temperature of the enclosure **100** is limited by the flammability limit of solvent of the coating on the glass article **302**, but generally ranges from 60° C. to 100° C.

[0065] The part carrier **300** continues to move the glass articles **302** along the manufacturing line path until it reaches the next manufacturing station or location, which, in embodiments, may be a curing apparatus **502**. In embodiments, the coating on each of the glass articles **302** is cured within the enclosure **100**, or the glass articles **302** may exit the enclosure **100** before being directed into the curing apparatus **502**. Thus, it is contemplated that the curing apparatus **502** can be positioned within the enclosure **100** or adjacent to the enclosure **100**, depending on the particular embodiment. The curing apparatus **502** may be any suitable type of curing apparatus, depending on the particular coating applied to the glass article **302**. For example, the curing apparatus **502** may be an oven or a light source (e.g., an infrared or UV light source). The glass article **302** can be removed from the enclosure **100** in a manner similar to the manner in which it was positioned within the enclosure **100**.

Examples

[0066] The examples are representative embodiments of the presently disclosed subject matter, and are not meant as limiting the scope of the claims.

[0067] Using ANSYS FLUENT™ (Ansys) software, a three-dimensional computational fluid dynamics (3D CFD) model was used to model the flow of fluid through an enclosure according to one or more embodiments discussed in the detailed description. Specifically, the modeled enclosure had a W.sub.inlet of 2.54 centimeters (cm), a W.sub.chamber of 7.62 cm, and two outlets with widths of 2.54 cm each. Additionally, W.sub.entry was 3.81 cm and the clearance between the gripping member and the entry port was 0.41 cm on each side of the gripping member. The velocity of air entering through the inlet was 1.8 m/s and the velocity of air exiting through the outlets was 1.0 m/s for each outlet; thus, there was an imbalance between the flow of air through the inlet and the outlets. The 3D CFD model was used to predict detailed flow patterns through the enclosure during steady state operation. The flow patterns of various fluids and particles are depicted by pathlines in FIGS. **6-10**.

[0068] FIG. **6** depicts pathlines generated by the 3D CFD model to visualize the flow of fluid from the inlet **104** through the enclosure **100**. As shown in FIG. **6**, all of the fluid that entered through the inlet **104** left the enclosure **100** through the outlets **109**. Additionally, the pathlines show that the flow of fluid was consistently in an upward direction, toward the top end **102** of the enclosure **100**. The pathlines do not depict swirling or eddies in the flow of fluid from the inlet **104** through the enclosure **100**, suggesting that the flow of fluid from the inlet **104** to the outlet **109** was substantially laminar.

[0069] FIG. **7** depicts pathlines emanating from the glass article **302**, which in this example is in the form of a glass vial. The pathlines show the flow of fluid from the surface of the glass article

302. This simulation accounted for the rotation of the glass article **302** within the enclosure **100** at a rate of 2000 RPM. Accordingly, the pathlines wrap around the glass article **302**. As in FIG. **6**, all of the pathlines in FIG. **7** point toward the outlets **109**, suggesting that none of the fluid in contact with the glass article **302** left the enclosure **100** through the entry port **108**.

[0070] FIG. **8** depicts the concentration of solvent vapor within the enclosure **100** when the concentration of solvent vapor at the surface of the glass article was 11% based on mass. FIG. **8** shows that the solvent vapor left the surface of the glass article and exited the enclosure **100** through the outlets **109**, and not through the entry port **108**.

[0071] FIG. **9** depicts the flow of ambient air into the enclosure **100** through the entry port **108**. In particular, the pathlines in FIG. **9** show that ambient air that entered the enclosure **100** through the entry port **108** exited the enclosure through the outlets **109**, and did not enter the chamber region **106** of the enclosure **100** or contact the glass article **302**. Additionally, the flow of ambient air into the enclosure **100** through the entry port **108** was believed to help prevent the leakage of solvent vapor from the enclosure through the entry port **108**.

[0072] Next, spherical particles having a diameter of 100 μm and a density of 2,000 kg/m.sup.3 were simulated entering the enclosure **100** through the entry port **108**. Pathlines depicted in FIG. **10** show that most of these particles were removed from the enclosure **100** through the outlets **109** before they entered the chamber region **106** of the enclosure **100**. However, some particles entered the chamber region **106** of the enclosure **100** and contacted the glass article **302**. It is believed that a disk or plate included on the part carrier, such as plate **303** (FIG. **3**), could catch these large particles before they entered the chamber region and could help direct smaller particles and air entering through the entry port **108** to the outlets **109** to be removed from the enclosure.

[0073] Additional modeling was performed on the flow of fluid through an enclosure **100** without a part carrier **300** or glass article **302**. Spherical particles having a diameter of 100 μm and a density of 2,000 kg/m.sup.3 were simulated entering the enclosure **100** through entry port **108**. FIG. **11** depicts pathlines which show that the particles entering the enclosure **100** through the entry port **108** and exiting the enclosure **100** through the outlets **109**. The particles did not enter the chamber region **106**, even when there is no gripping member **301** occupying space within the entry port **108**. Thus, it is unlikely that contaminants having a diameter of less than 100 μm and a density of 2,000 kg/m.sup.3 would enter the enclosure **100** through the entry port **108** in the space between gripping members **301**.

[0074] In a first aspect of the present disclosure, an enclosure for providing a controlled environment includes a central plane extending through a top end of the enclosure and a bottom end of the enclosure and bisecting the enclosure along a width of the enclosure; an inlet at the bottom end of the enclosure having an inlet width, W.sub.inlet; an enclosure wall extending from the inlet to the top end of the enclosure; an entry port at the top end of the enclosure configured to receive a part carrier; and an outlet between the entry port and the chamber region of the enclosure wall. The enclosure wall includes a chamber region and a transition region between the inlet and the chamber region. A width of the chamber region, W.sub.chamber, is substantially constant through the chamber region. The width of the enclosure in the transition region decreases from W.sub.chamber to W.sub.inlet, and a ratio of W.sub.inlet to W.sub.chamber may be from 1:2 to 1:5. The central plane passes through the inlet and the entry port of the enclosure. The outlet extends along an outlet axis that is oriented at a non-zero angle with respect to the central plane.

[0075] A second aspect of the present disclosure may include the first aspect where the enclosure comprises reflection symmetry with respect to the central plane.

[0076] A third aspect of the present disclosure may include either of the first or second aspects were the width of the enclosure transitions from W.sub.inlet to W.sub.chamber over a distance of from 200 mm to 900 mm.

[0077] A fourth aspect of the present disclosure may include any of the first through third aspects where W.sub.inlet is from 4 mm to 45 mm.

[0078] A fifth aspect of the present disclosure may include any of the first through fourth aspects were W.sub.chamber is from 20 mm to 90 mm.
[0079] A sixth aspect of the present disclosure may include any of the first through fifth aspects where the enclosure wall comprises an S-shaped curve with an inflection point within the transition region.
[0080] A seventh aspect of the present disclosure may include any of the first through sixth aspects were the outlet axis is normal with respect to the central plane.
[0081] In an eighth aspect of the present disclosure, a manufacturing line for producing glass articles includes an enclosure and a part carrier. The enclosure includes a central plane extending through a top end of the enclosure and a bottom end of the enclosure and bisecting the enclosure along a width of the enclosure; an inlet at the bottom end of the enclosure having an inlet width, W.sub.inlet; an enclosure wall extending from the inlet to the top end of the enclosure; an entry port at the top end of the enclosure configured to receive the part carrier; and an outlet between the entry port and the chamber region of the enclosure wall. The enclosure wall includes a chamber region and a transition region between the inlet and the chamber region. A width of the chamber region, W.sub.chamber, is substantially constant through the chamber region. The width of the enclosure in the transition region decreases from W.sub.chamber to W.sub.inlet, and a ratio of W.sub.inlet to W.sub.chamber may be from 1:2 to 1:5. The central plane passes through the inlet and the entry port of the enclosure. The outlet extends along an outlet axis that is oriented at a non-zero angle with respect to the central plane. A gripping member of the part carrier is positioned through the entry port, and the part carrier is configured to move a glass article through the chamber region of the enclosure.
[0082] A ninth aspect of the present disclosure may include the eighth aspect where the part carrier comprises a plate positioned between the entry port and the chamber region and extending along a plane normal to the central plane, the gripping member extending through the plate.
[0083] A tenth aspect of the present disclosure may include the ninth aspect where the plate has a width, W.sub.plate, that is greater than or equal to W.sub.chamber.
[0084] An eleventh aspect of the present disclosure may include either of the ninth or tenth aspects where the entry port has a width, W.sub.entry, and the width of the plate W.sub.plate is greater than the width of the entry port, W.sub.entry.
[0085] A twelfth aspect of the present disclosure may include any of the ninth through eleventh aspects where the width of the entry port, W.sub.entry, is less than the width of the chamber region, W.sub.chamber.
[0086] A thirteenth aspect of the present disclosure may include any of the ninth through twelfth aspects where the plate extends into the outlet.
[0087] A fourteenth aspect of the present disclosure may include the ninth aspect where the plate comprises a disk with a diameter greater than or equal to W.sub.chamber.
[0088] A fifteenth aspect of the present disclosure may include any of the eighth through fourteenth aspects where outlet axis is normal with respect to the central plane.
[0089] In a sixteenth aspect of the present disclosure, a method for transporting a coated article comprises positioning the coated article within an enclosure; supplying a flow of fluid to the enclosure through the inlet; removing a flow of fluid from the enclosure through the outlet; and moving the coated article along a path through the enclosure, where the path is substantially parallel to the central plane. The enclosure comprises a central plane extending through a top end of the enclosure and a bottom end of the enclosure and bisecting the enclosure along a width of the enclosure; an inlet at the bottom end of the enclosure having an inlet width, W.sub.inlet; an enclosure wall extending from the inlet to the top end of the enclosure; an entry port at the top end of the enclosure configured to receive a part carrier; and an outlet between the entry port and the chamber region of the enclosure wall. The enclosure wall comprises a chamber region and a transition region between the inlet and the chamber region. A width of the chamber region,

W.sub.chamber, is substantially constant through the chamber region. The width of the enclosure in the transition region decreases from W.sub.chamber to W.sub.inlet, and a ratio of W.sub.inlet to W.sub.chamber is from 1:2 to 1:5. The central plane passes through the inlet and the entry port of the enclosure and the outlet extends along an outlet axis oriented at a non-zero angle with respect to the central plane.

[0090] A seventeenth aspect of the present disclosure may include the sixteenth aspect where vapors evaporate from the coated article during the moving of the coated article through the enclosure and are extracted from the enclosure through the outlet.

[0091] An eighteenth aspect of the present disclosure may include either of the sixteenth or seventeenth aspects where moving the coated article further comprises rotating the coated article around an axis central to the coated article and substantially parallel to the central plane.

[0092] A nineteenth aspect of the present disclosure may include the eighteenth aspect where the rotation of the coated article may be at a rate from 1000 to 3000 rpm.

[0093] A twentieth aspect of the present disclosure may include any of the sixteenth through eighteenth aspects where a pressure within the enclosure is less than an ambient air pressure.

[0094] A twenty-first aspect of the present disclosure may include any of the sixteenth through twentieth aspects where removing the flow of fluid from the enclosure comprises applying a vacuum to the outlet.

[0095] A twenty-second aspect of the present disclosure may include any of the sixteenth through twenty-first aspects where the flow of fluid through the enclosure is substantially laminar.

[0096] A twenty-third aspect of the present disclosure may include any of the sixteenth through twenty-second aspects where the fluid supplied to the enclosure has a temperature from 20 to 25° C. and a relative humidity of less than 60%.

[0097] A twenty-fourth aspect of the present disclosure may include any of the sixteenth through twenty-third aspects where supplying the flow of fluid to the enclosure further comprises passing the air through a HEPA filter.

[0098] A twenty-fifth aspect of the present disclosure may include any of the sixteenth through twenty-fourth aspects where the fluid supplied to the enclosure has a temperature greater than 300° C. such that the coated article is cured within the enclosure.

[0099] It will be apparent to those skilled in the art that various modifications and variations can be made to the embodiments described herein without departing from the spirit and scope of the embodiments. Since modifications, combinations, sub-combinations, and variations of the disclosed embodiments incorporating the spirit and substance of the embodiments may occur to persons skilled in the art, the embodiments should be construed to include everything within the scope of the appended claims and their equivalents.

Claims

1. A method for transporting a coated article, the method comprising: positioning the coated article within an enclosure, the enclosure comprising: a central plane extending through a top end of the enclosure and a bottom end of the enclosure and bisecting the enclosure along a width of the enclosure; an inlet at the bottom end of the enclosure having an inlet width, W.sub.inlet; an enclosure wall extending from the inlet to the top end of the enclosure, the enclosure wall comprising a chamber region and a transition region between the inlet and the chamber region, wherein a width of the chamber region, W.sub.chamber, is substantially constant through the chamber region, the width of the enclosure in the transition region decreases from W.sub.chamber to W.sub.inlet, and a ratio of W.sub.inlet to W.sub.chamber is from 1:2 to 1:5; an entry port at the top end of the enclosure configured to receive a part carrier, wherein the part carrier is configured to move the coated article through the chamber region; and an outlet between the entry port and the chamber region of the enclosure wall; wherein the central plane passes through the inlet and the entry port of the enclosure and the outlet extends along an outlet axis oriented at a non-zero angle with respect to the central plane; supplying a flow of fluid to the enclosure through the inlet; removing a flow of fluid from the enclosure through the outlet; and moving the coated article along a path through the enclosure, wherein the path is substantially parallel to the central plane.

2. The method of claim 1, wherein vapors evaporate from the coated article during the moving of the coated article through the enclosure and are extracted from the enclosure through the outlet.

3. The method of claim 1, wherein moving the coated article further comprises rotating the coated article around an axis central to the coated article and substantially parallel to the central plane.

4. The method of claim 3, wherein the rotation of the coated article is at a rate from 1000 to 3000 rpm.

5. The method of claim 1, wherein a pressure within the enclosure is less than an ambient air pressure.

6. The method of claim 1, wherein removing the flow of fluid from the enclosure comprises applying a vacuum to the outlet.

7. The method of claim 1, wherein the flow of fluid through the enclosure is substantially laminar.

8. The method of claim 1, wherein the fluid supplied to the enclosure has a temperature from 20 to 25° C. and a relative humidity of less than 60%.

9. The method of claim 1, wherein supplying the flow of fluid to the enclosure further comprises passing the air through a HEPA filter.

10. The method of claim 1, wherein the fluid supplied to the enclosure has a temperature greater than or equal to 300° C.

11. The method of claim 2, wherein the vapors are directed from the outlet, through a manifold and a vacuum source, and to a solvent recovery system that separates the vapors from the rest of the fluid that is removed from the outlet.

12. The method of claim 1, wherein a temperature within the enclosure is between 60° C. and 100° C.

13. The method of claim 1, further comprising curing the coated article in the enclosure.

14. The method of claim 1, wherein the width of the enclosure transitions from W.sub.inlet to W chamber over a distance of from 200 mm to 900 mm.

15. The method of claim 1, wherein W.sub.inlet is from 4 mm to 45 mm.

16. The method of claim 1, wherein W.sub.chamber is from 20 mm to 90 mm.

17. The method of claim 1, wherein the enclosure comprises reflection symmetry with respect to the central plane.

18. The method of claim 1, wherein the outlet axis is normal with respect to the central plane.

19. The method of claim 1, wherein the enclosure wall comprises an S-shaped curve with an inflection point within the transition region.